

Lab 4

FTP MachLe MSE
FS 2026

Feature Engineering

Machine Learning
WÜRC

«*Feature engineering is the process of transforming raw data into features that better represent the underlying problem to the predictive models, resulting in improved model accuracy on unseen data.*»DR. JASON BROWNLEE

«*At the end of the day, some machine learning projects succeed and some fail. What makes the difference? Easily the most important factor is the features used.* »PROF. PEDRO DOMINGOS, UNIV. OF WASHINGTON ¹

After this unit, you are able ...

Lernziele/Kompetenzen

- to *classify* different datatypes: numerical, categorical, text, images, sound,...
- to *preprocess* data that contain *missing values*, you can *convert* categorical variables into numbers and you can *scale* and *transform* data appropriate to your application.
- to *scale* data using `sklearn.preprocessing`, namely `MinMaxScaler`, `MaxAbsScaler`, `StandardScaler`, `RobustScaler`, `Normalizer`, `QuantileTransformer`. You are able to explain the working principle of these scalers and know when to use them.
- to *binarize* categorical data by one-hot-encoding using either `pandas.get_dummies` or `sklearn.preprocessing.OneHotEncoder`.
- to apply *binning* to a continuous dimension for better prediction using `numpy.digitize`.
- to *select* the most relevant features using variance threshold (ANOVA) and the feature importance based on the Gini index from decision trees (automatic feature selection).
- to *generate new features* by combination of features (polynomial feature generation using `sklearn.preprocessing.PolynomialFeatures`), especially for linear models.
- to *generate meaningful features out of text data* (bag-of-words, n-grams, tf-idf) using `TfidfVectorizer` from `sklearn.feature_extraction.text`.
- Generate *time, frequency and time-frequency domain features* of raw audio signals to *classify sound*.

¹A Few Useful Things to Know about Machine Learning: <https://homes.cs.washington.edu/~pedrod/papers/cacm12.pdf>

1. Prediction of City Bike usage [A,II]

The solution Jupyter notebook can be found on moodle:

`Lab4_A1_CitibikeUsage_solution.ipynb`

2. Automatic feature selection using variance threshold (ANOVA) and trees [A,II]

The solution Jupyter notebook can be found on moodle:

`Lab4_A2_AutomaticFeatureSelectionANOVA_solution.ipynb`

3. Yelp Business Classification using Bag-of-Words and tf-idf [A,III]

The solution Jupyter notebook can be found on moodle:

`Lab4_A3_YelpBusinessClass_BoW_tf-idf_solution.ipynb`

4. Free Spoken Digits Classification (optional) [A,III]

The solution Jupyter notebook can be found on moodle:

`Lab4_A4_FSDD_classification_SOLUTION.ipynb`